

APPENDIX 4.1: ECOLOGICAL REGENERATION, BIOSPHERIC PARITY, AND TSB GRANT CIRCUIT

1. Fundamental Economic Mechanism of the Reserve Fund (S_reserve)

The Transceptor Global ecosystem algorithmically links any economic act and computing power with the physical factor of biosphere regeneration:

- Upon any emission of T_net and T_nat tokens, fuel burning, or infrastructure operation, the clearing gateway retains the Entropy Deduction Index (K_entropy), calculated based on Life Cycle Assessment (LCA) of the equipment.
- The exchange rate spread upon converting national sub-tokens to global T_net is forcibly extracted and deposited into the Regeneration Reserve Fund (S_reserve).
- From every 2.50% commission split of TSB-Clearing, exactly 20% (0.50% of the transaction volume) is directed to the S_reserve fund according to the approved treasury Waterfall model. Additionally, up to 0.5% of transit and resource royalties are retained prior to allocating any other budgets.
- **Treasury Invariant and Escrow:** Incoming funds (minus the entry split) are deposited into sovereign securities (T-Bills) with 100% principal protection (Zero-Burn Model) for a term of up to 3 years in strict compliance with Appendices 1.0, 1.2, and 1.5.
- **Incubation Sprint Invariant (Phase 0 — up to 24 months):** Algorithmic allocation of funds and grant payouts from the S_reserve fund are activated upon completion of Phase 0 (0–24 months) — after the commercial release of the platform, deployment of the Digital Agora, Scientific Guilds, and the IoT sensor network.

2. Grant Support for Science and De-extinction Programs

Funds from the S_reserve fund are allocated across the following priority areas:

- **De-extinction and Fauna Protection:** Targeted funding for genetic research, laboratories, and biotechnology centers working to restore lost or endangered species (including endemics of Australia, New Zealand, Madagascar, and oceanic islands).
- **Hydrobiont Replenishment:** Funding for programs to restore fish populations, coral reefs, and mangrove ecosystems affected by industrial fishing.
- **Reserves and Eco-Corridors:** Acquisition and withdrawal of land from industrial use to create continuous animal migration routes, as well as the deployment of IoT networks, LiDAR scanning, and hydro-oracles to protect against poaching and deforestation.
- **B2G Strategic Fork (S_reserve^nat -> S_reserve^global):** 100% of National Eco-Fund assets are offered as grants to local researchers and contractors. In the absence of verified projects from sovereign authorities within 12–24 months, capital is automatically transferred to the Global Eco-Fund for international biosphere programs.

3. Payout Regulations via PoUCC and Tranche Staging

1. **Researcher Registration:** Scientists, biologists, and ecologists establish their subjectivity via W3C DID passports.
2. **Contribution Accounting:** Live man-hours (H_live_effective) and cognitive work are accounted for under the PoUCC standard with the accrual of quants (+0.01) to the Competence Index (delta I_cp) and Social Contribution Index (delta I_sc), taking into account utility limits under Appendix 1.3.
3. **Phased Funding (Tranche Staging):** Grant tranche releases operate under the Proof-of-Useful-Outcome (PoUO) principle — liquidity in T_net tokens is released in stages from TSB escrow pools upon confirmation of anti-entropy effects by IoT sensors audited by Scientific Guilds.